Technische Hochschule Augsburg

STATISTICS

International Information Systems, 3rd semester

(8) Test Statistics (1)

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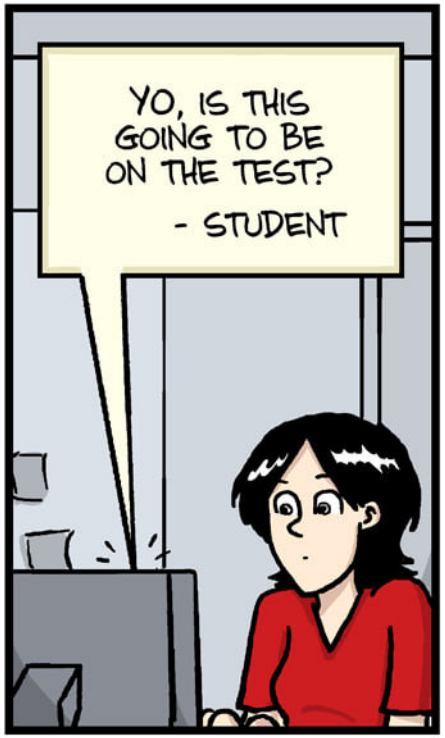
https://phdcomics.com/comics/archive_print.php?comicid=1873

DON'T FORGET TO REGISTER FOR THE EXAM!

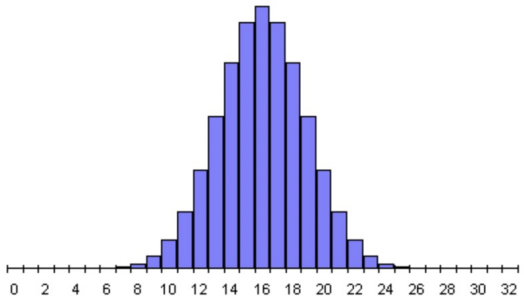


- Register for the exam by Nov 17th!
- Make sure you register for the right "Statistics" exam
  - Exam nr.: 9772010 (SPO2021) or 9872010 (SPO2024)

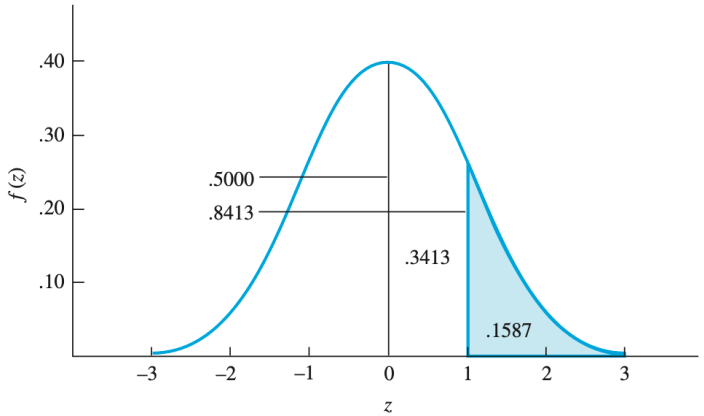
LAST WEEK

Towards Inferential Statistics

- **From Distributions to Probabilities**
  - We observe the **frequency distribution** of values in the sample, we want to know about their **probability** in the population
  - We'll start with an **assumption of normality**

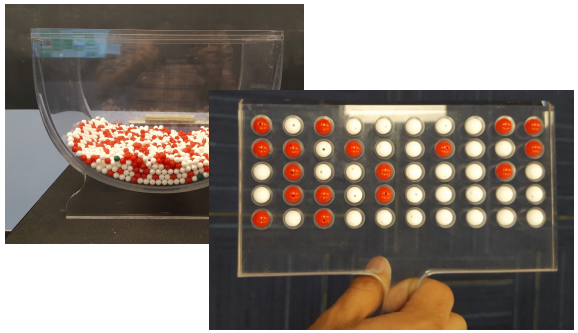


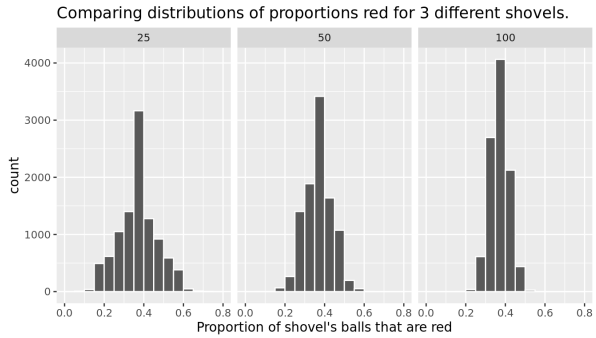
- **Normal Distributions,** defined by μ and σ
  - If the **Area under the Curve** is 1, then we can interpret the distribution in terms of ***probability of the values*** that a variable can take
  - If you randomly drew a score from this distribution, which part of the curve will the score come from?



LAST WEEK

Towards Inferential Statistics





- **Sampling**
  - **Sampling Distribution:** distribution of a sample statistic in samples from the same population
  - **Central Limit Theorem**: given a population with mean μ and variance σ^2, the sampling distribution of the mean for a sample size N
    - will have a mean equal to μ and a variance equal to $\frac{\sigma^2}{N}$
    - will ***approximate*** a Normal Distribution as N increases: **our guesses (based on the samples) get more consistent and precise (less likely due to chance) as N increases**

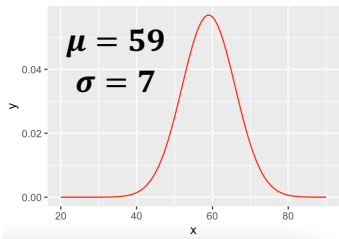
- **Hypothesis Testing:** we usually contrast two hypotheses
  - H_1 **or Alternative Hypothesis**: *an effect is present*
  - H_0 **or Null Hypothesis**: *an effect is absent* (what we observed is due to chance)
    - **The Null Hypothesis is assumed true until we manage to reject it**

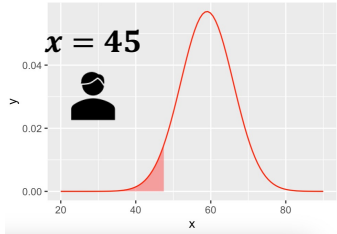
$$P(\text{result} \mid H_0)$$

LAST WEEK

Towards Inferential Statistics

- **Using the Normal Distribution for Hypothesis Testing: single scores**
  - $\boldsymbol{H_1}$: *a score does not belong to this population*
  - $\boldsymbol{H_0}$: *a score does belong to this population*





1. How does the **population** look like?

2. Given a single score, how likely is it that **a value as high / as low as this** would be randomly picked **from this population**?
   - **low probability**: reject H_0
   - **not so low probability**: we fail to reject H_0

3. It's the area below the curve for $\boldsymbol{x = 45}$ in the normal distribution with $\boldsymbol{\mu = 59}$ and $\boldsymbol{\sigma = 7}$

$$P(\text{score_this_low}|H_0) = 0.0228$$

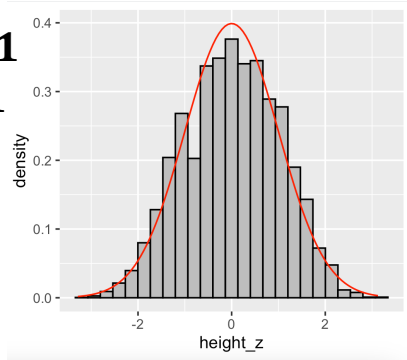
LAST WEEK

Towards Inferential Statistics

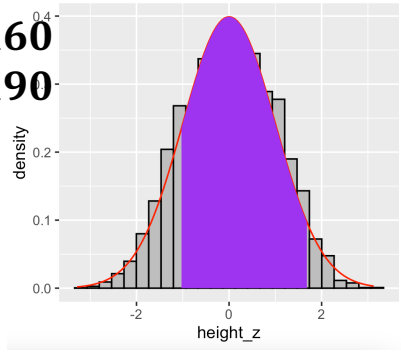
- **Last week: R_exercise_9.R** and **students.csv**
- **Homework**: how likely would you say it is that a student is between 160 and 190 cm tall?
  - **Solution: R_ exercise_9_solution.R**

1. How does the **population** look like?
2. How likely is it that **a value in this range** would be randomly picked **from this population**?
3. It's the area below the curve for $x > 160$ and $x < 190$ in the normal distribution with $\mu = 171$ and $\sigma = 11$

$\mu = 171$
$\sigma = 11$



$x > 160$
$x < 190$



$$P = 0.80$$

Questions?

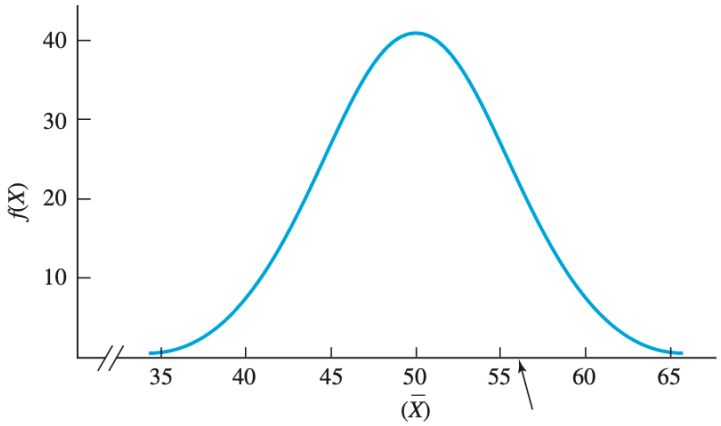
PRELIMINARY COURSE PLAN

- Introduction, Descriptive statistics: measures of central tendency
- Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations
- Basic probability theory
- Introduction to inferential statistics: sampling, hypothesis testing
- **[Today]** Test Statistics (I)
- [Nov 13th] **No class *(Tag der Lehre)***
- [Nov 20th-Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

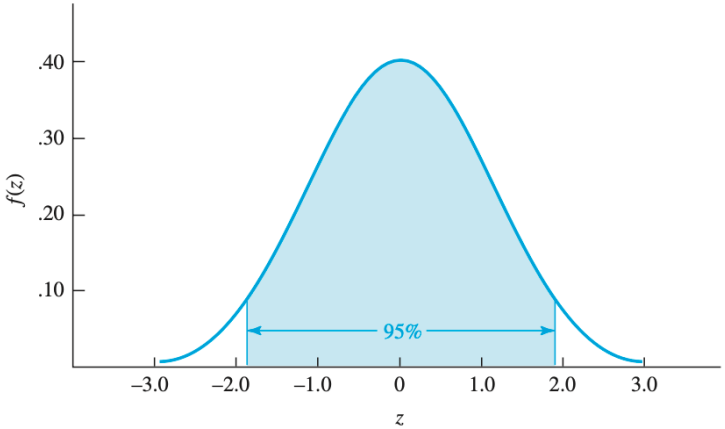
TODAY'S TOPICS AND LEARNING GOALS

- **Tests of the Mean: One Sample**
  - Given a population, calculate the **95% Probable Limits** of the scores from the population
  - Given a population, calculate the **95% Confidence Interval** of the samples from the sampling distribution
  - Calculate the probability of the Null Hypothesis that **a sample** comes from **a sampling distribution**
- **Type 1 and Type 2 Error**
  - Understand and explain the difference between **Type 1 and Type 2 Error** and between α and β
  - Explain the consequences for a **smaller or bigger α**

TESTS OF THE MEAN: ONE SAMPLE



PROBABLE LIMITS

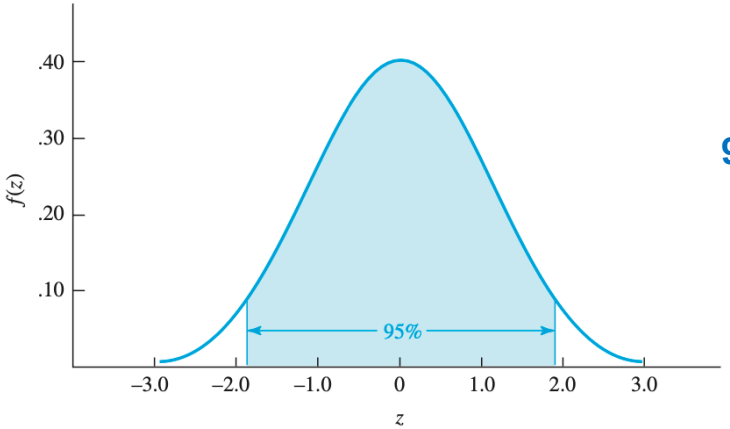


- We can interpret the area under the curve of a Standard Normal Distribution as a probability
  - Probability that a score will fall in a certain range
- If we look at a Standard Normal Distribution ($\mu = 0,\ \sigma = 1$), we can talk about **probability of z-scores**

$$z = \frac{\text{score} - \text{mean}}{SD}$$

- We know that a z-score of **1.96** is the **cutoff value** delimiting **95% of the scores** in a Standard Normal Distribution

PROBABLE LIMITS



95% Probable Limits, where $\frac{\text{score} - \text{mean}}{SD} = \pm 1.96$

- We know that a z-score of **1.96** is the **cutoff value** delimiting **95% of the scores** in a Standard Normal Distribution

- For a 95% probability:
  - lower limit = -1.96 SD from the mean
    - only 2.5% of the datapoints are smaller than this
  - upper limit = 1.96 SD from the mean
    - only 2.5% of the datapoints are bigger than this

$$\text{lower_limit} = \text{mean} - (1.96 * SD)$$

$$\text{upper_limit} = \text{mean} + (1.96 * SD)$$

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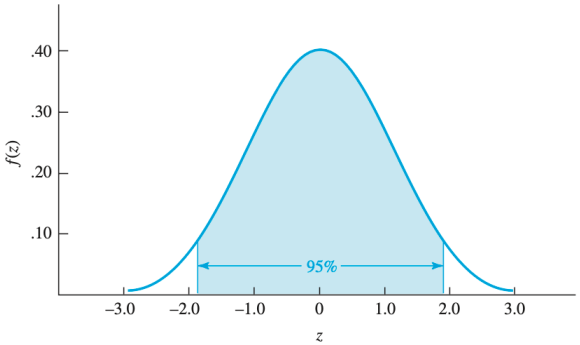
EXERCISE



Study on the effect of stress on children behavior (Williamson 2008)

- Scores obtained with a behavioral checklist, the anxious/depressed subscale of Achenbach's Youth Self-Report Inventory (YSR)
  - Non-stressed children: mean score of 50 (SD = 10)

In which range will most of the scores (95%) for non-stressed children fall?

R_exercise_12.R

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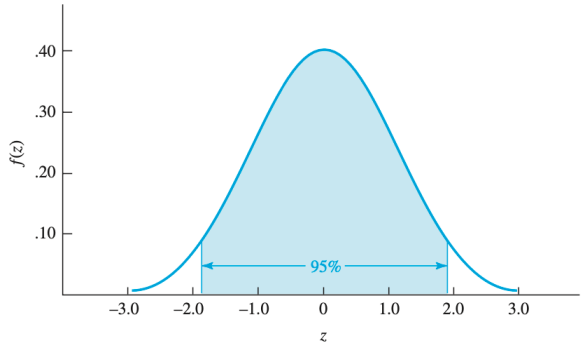
EXERCISE



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  - Non-stressed children: mean score of 50 (SD = 10)

In which range will most of the scores (95%) for non-stressed children fall?

- $\text{lower_limit} = \text{mean} - (1.96 * SD)$
- $\text{upper_limit} = \text{mean} + (1.96 * SD)$

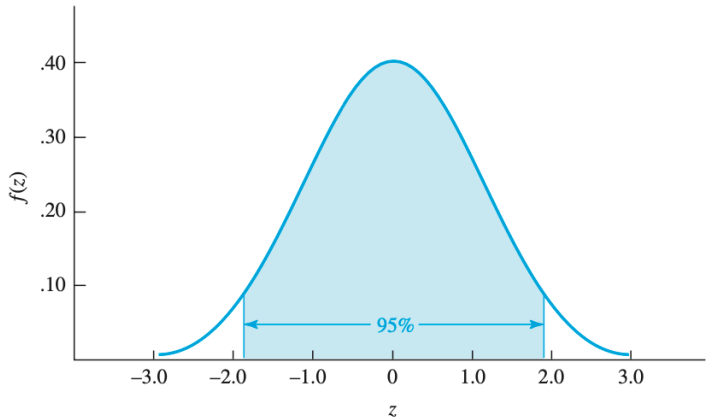
$$\text{lower_limit} = 50 - (1.96 * 10) = 30.4$$

$$\text{upper_limit} = 50 + (1.96 * 10) = 69.6$$

If we draw a child at random from this population, 95% of the time her score will lie between 30.4 and 69.6

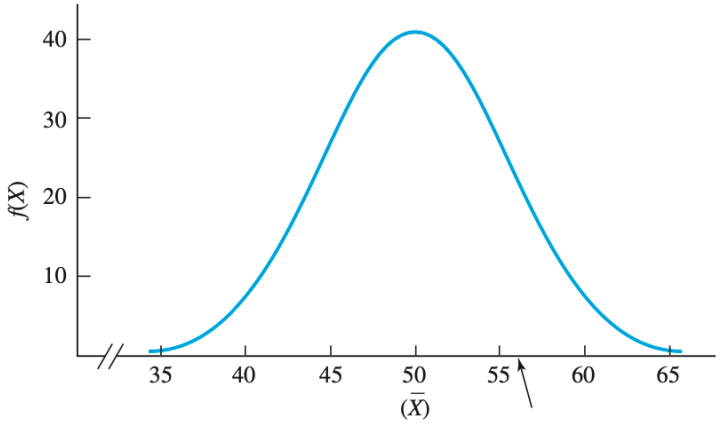
FROM A DISTRIBUTION TO A SAMPLING DISTRIBUTION

Distribution of scores



Where would we find most scores?
How probable is a score this high / low for this distribution?

Distribution of samples



Where would we find most samples?
How probable is a sample with a mean this high / low for this sampling distribution?

THE SAMPLING DISTRIBUTION

μ of sampling distribution:
μ of the population

Variance of sampling distribution:

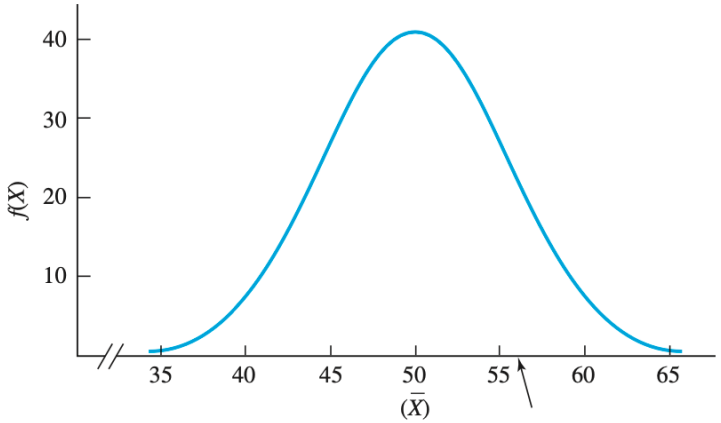
$$\frac{\sigma^2}{N}$$

Standard deviation of sampling distribution: $\sigma_{\bar{X}}$
also known as ***Standard Error (SE)***

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{N}}$$

N: sample size

Distribution of samples



CONFIDENCE LIMITS AND CONFIDENCE INTERVALS

- When trying to understand who will win an election, we can guess what percentage of votes one candidate will get
  - e.g. “between 30 and 40%” (Confidence Interval, CI)

- Our useful is our guess?
  It depends on ***how spread out*** the data is, ***how big a sample*** we can observe, and ***how sure*** we want to be about our guess
  - If the **size of our sample** allows us to say that
    “we can be **95% sure** that **our candidate** will get between 30 and 70% of the votes”,
    how informative / useful is this guess?
  - With a **bigger sample**, we may be able to say that
    “we can be **95% sure** that **our candidate** will get between 40 and 50% of the votes”,

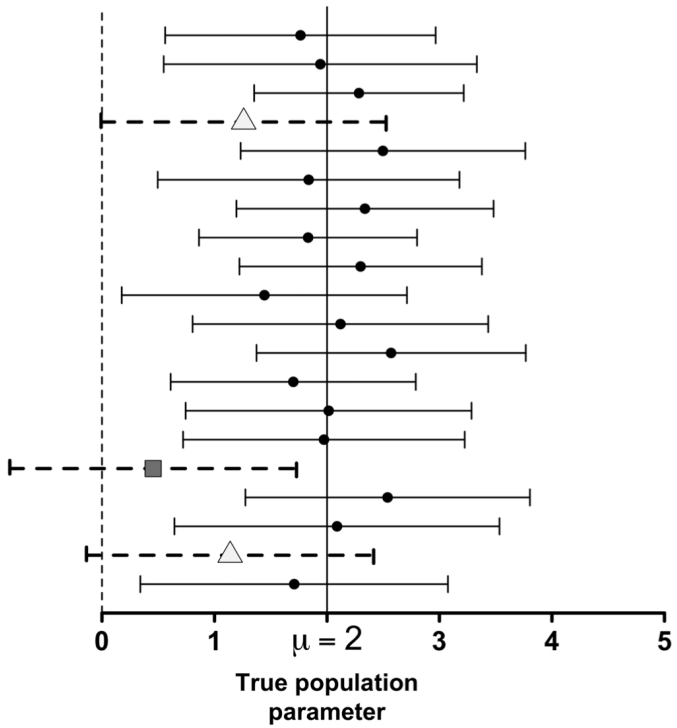
Given the same level of certainty (95%)
and the same data (same spread),
the sample size makes a big difference between
a wide interval and a narrower one

CONFIDENCE LIMITS AND CONFIDENCE INTERVALS

- As we know, a sample parameter (e.g. the mean) may be very different than the population parameter

- Given a sample parameter, we can compute the **Confidence Interval** (**CI**) which is an estimate of where we are likely to find the population parameter

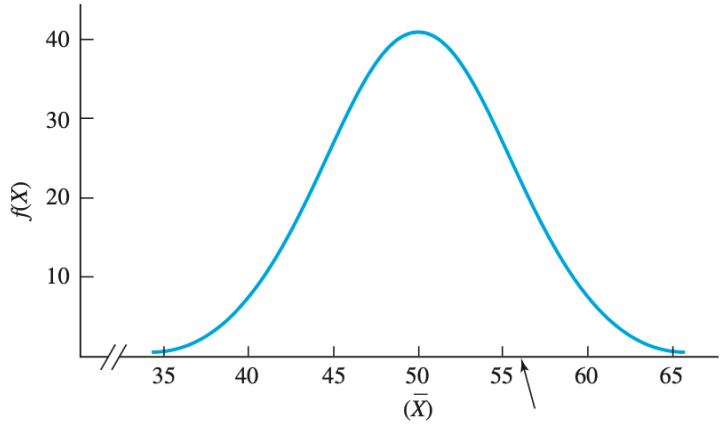
"the Dance of the confidence intervals"
(Cumming 2012, 2014)
Samples of 60 data points from
the same distribution ($\mu = 2, \sigma = 5$)



CONFIDENCE LIMITS AND CONFIDENCE INTERVALS

95% Confidence Intervals (CI)

$$\frac{\text{score} - \text{mean}}{SE} = \pm 1.96$$



- In a Sampling Distribution, **95% of the samples** will be **1.96 SE** from the mean (**Confidence Interval**)
- With large sample sizes, we can use the mean of the sample to find the **Confidence Limits**
- For a 95% probability:
  - lower limit = -1.96 SE from the mean
    - only 2.5% of the times the population mean will be smaller
  - upper limit = 1.96 SE from the mean
    - only 2.5% of the times the population mean will be bigger

$$\text{lower_limit} = \text{sample_mean} - (1.96 * SE)$$

$$\text{upper_limit} = \text{sample_mean} + (1.96 * SE)$$

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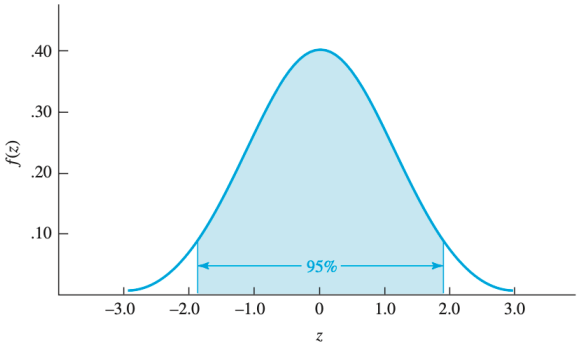
USING THE SAMPLING DISTRIBUTION FOR HYPOTHESIS TESTING

Mean of a sample



Scores obtained with a behavioral checklist,
the anxious/depressed subscale of Achenbach's Youth Self-Report Inventory (YSR)

- Population of non-stressed children: mean score of 50 (SD = 10)

  - If μ and σ are known, we could test a **hypothesis about a sample** too
    - We have just tested a sample of 5 children with a mean score of 56 ($\bar{X} = 56$)
    - What could our Alternative Hypothesis (H_1) and our Null Hypothesis (H_0) be?
      - H_1 : they are a random sample from a ***different population***
      - H_0: they are a random sample from a ***population of non-stressed children***

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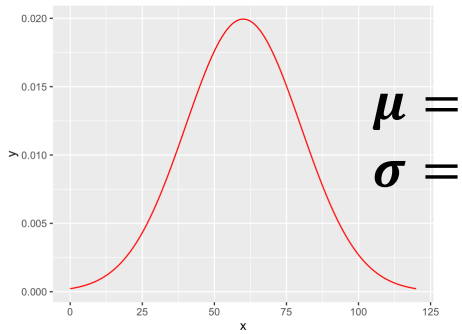
USING THE SAMPLING DISTRIBUTION FOR HYPOTHESIS TESTING

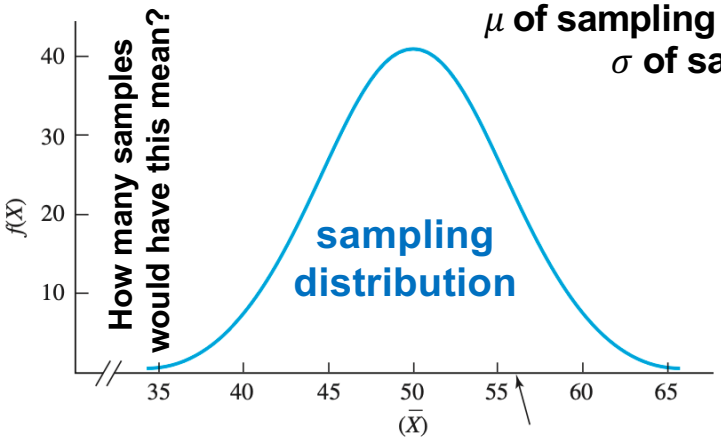
Mean of a sample



$\mu = 50$
$\sigma = 10$

1. How does the population of non-stressed children look like?
   We know the size and mean of the population
2. We know the size and the mean of our sample
3. How likely is it that, if we picked **a random sample** from the population, their mean would be ***as high as 56?***

$N = 5$
$\bar{X} = 56$



mean of behavior issue scores in a sample

μ of sampling distribution: μ of the population
σ of sampling distribution: $\sigma_{\bar{X}}$

↗ We need to compute the σ (SD) of the sampling distribution, also known as **Standard Error**

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{N}}$$

N: sample size

Big sample size → small variation!

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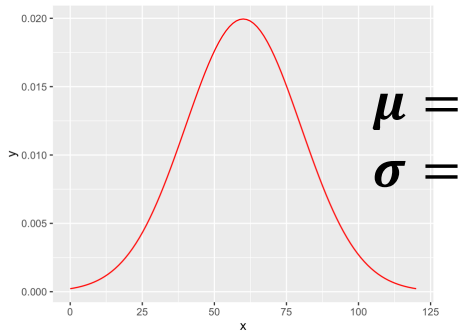
R_exercise_13.R

USING THE SAMPLING DISTRIBUTION FOR HYPOTHESIS TESTING

Mean of a sample



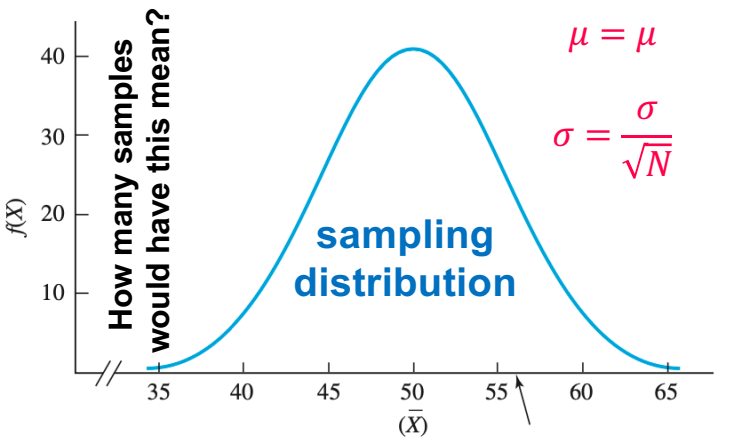
$\mu = 50$

$\sigma = 10$

1. How does the population of non-stressed children look like?
   We know the size and mean of the population
2. We know the size and the mean of our sample
3. How likely is it that, if we picked **a random sample** from the population, their mean would be ***as low as 56?***

$N = 5$

$\bar{X} = 56$



mean of behavior issue scores in a sample

- Let's say for us a probability of p < 0.05 is low enough to reject the Null Hypothesis
  - **low probability** ($p < \alpha$): reject H_0
  - **not so low probability** ($p \geq \alpha$): we fail to reject H_0

$\alpha = 0.05$

- The probability of a sample as low as 56 is the area below the curve for $\bar{X} = 56$ in the **sampling distribution** with $\mu = 50$ and $\sigma = \frac{10}{\sqrt{5}} = 4.47$

$P(\text{sample}|H_0) = 0.0901$

We can't reject the Null Hypothesis!

Questions?

TYPE 1 AND TYPE 2 ERROR

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TRUE UNTIL REJECTED

When we formulate our hypotheses,
the Null Hypothesis is assumed ***true until rejected***

- We reject the Null Hypothesis if
  - the ***evidence*** is ***too inconsistent with this belief***
  - the probability of the Null Hypothesis being true **given this evidence** is very **low**

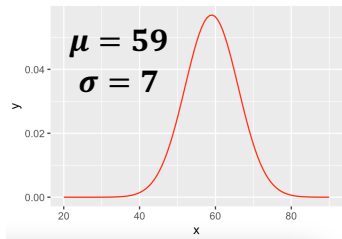
$$P(\text{result} \mid H_0)$$

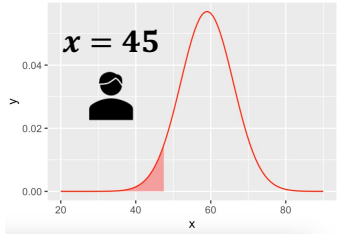
- **very low:** we ***reject*** the Null Hypothesis
- **not low enough:** we ***fail to reject*** the Null Hypothesis

How low is low enough?

HYPOTHESIS TESTING FOR A SINGLE SCORE USING THE NORMAL DISTRIBUTION

- **Using the Normal Distribution for Hypothesis Testing: single scores**
  - $\boldsymbol{H_1}$: *a score does not belong to this population*
  - $\boldsymbol{H_0}$: *a score does belong to this population*





1. How does the **population** look like?

2. Given a single score, how likely is it that **a value as high / as low as this** would be randomly picked **from this population**?
   - **low probability**: reject H_0
   - **not so low probability**: we fail to reject H_0

3. It's the area below the curve for $\boldsymbol{x = 45}$ in the normal distribution with $\boldsymbol{\mu = 59}$ and $\boldsymbol{\sigma = 7}$

$$P(\text{score_this_low}|H_0) = .0228$$

Is this low enough?

HOW DO WE DECIDE?

Type 1 Error

	H_0 is true	H_0 is false
fail to reject H_0		Type II error
reject H_0	Type I error	

- When we make a decision about rejecting or failing to reject H_0 we could of course be wrong
  - **Type 1 Error** indicates the error of rejecting H_0 ***when we shouldn't have***
  - Before running our inferential statistics tests, we need to decide on what probability of Type I error we can live with

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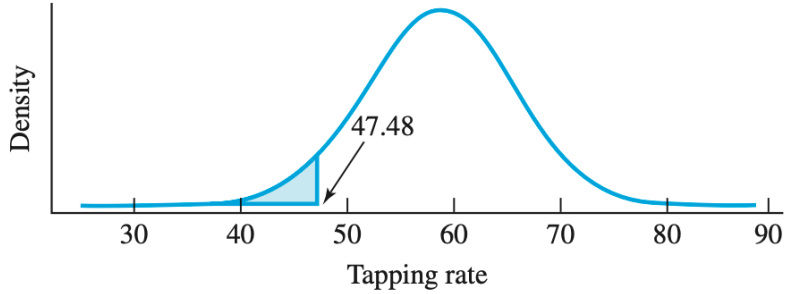
HOW DO WE DECIDE?

Type 1 Error and the critical value

	H_0 is true	H_0 is false
fail to reject H_0		Type II error
reject H_0	Type I error	

$\alpha = 5\%$



- The probability of Type 1 error ***we can live with*** is called α or **critical value**
  - the value of a statistic ***beyond which*** we decide to reject H_0

Shall I just be on the safe side and pick a very small α so we don't falsely reject H_0? … it depends!

  - what α you choose depends on ***what*** you are using inferential statistics ***for*** (what risk of Type 1 error are you ok with?)

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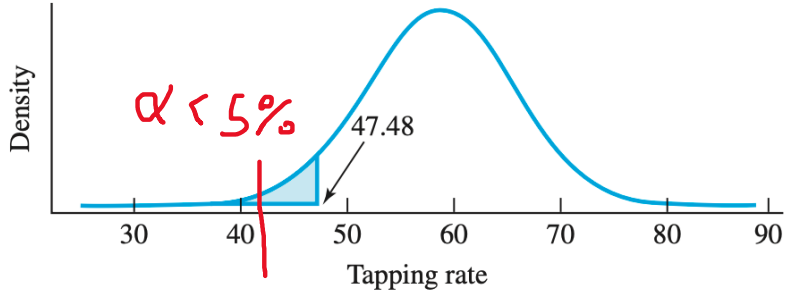
HOW DO WE DECIDE?

Type 2 Error

	H_0 is true	H_0 is false
fail to reject H_0		Type II error
reject H_0	Type I error	

$\alpha = 5\%$



- **Type 2 Error** indicates the error of failing to reject H_0 ***when we should have***
  - a stricter α (lower risk of Type 1 error) means a higher risk of Type 2 error
  - if we are too "careful", then we may lose the opportunity to reject H_0

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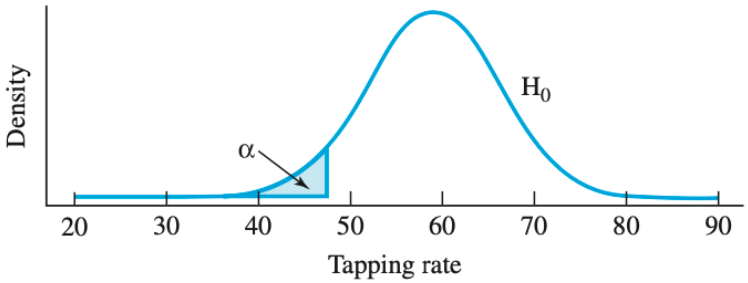
"healthy adults" H_0

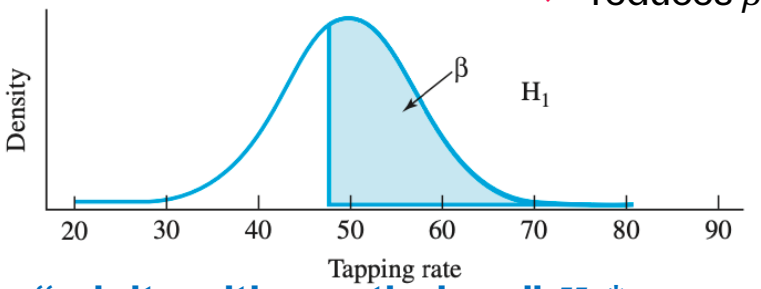


smaller α:

- ↗ reduces α, chances of false positives
- ↗ increases chances of **Type 2 Error** (we should reject H_0 but we can't) – that is β

bigger α:

- ↗ increases chances of **Type I Error** (falsely) rejecting H_0
- ↗ reduces β, chances of false negatives



"adults with a pathology" H_1*

* We almost never know the distribution for H_1

ONE VS. TWO-TAILED HYPOTHESES

Our research hypotheses (H_1) can be one- or two-tailed

- **Directional or one-tailed**: states the direction of the effect
  'students will know more about statistics after this class'

- **Non-directional or two-tailed**: an effect will occur, but it doesn't state the direction
  'students' knowledge of statistics will change after this class'

ONE VS. TWO-TAILED HYPOTHESES

If we expect someone tapping too slowly not to belong to the healthy population

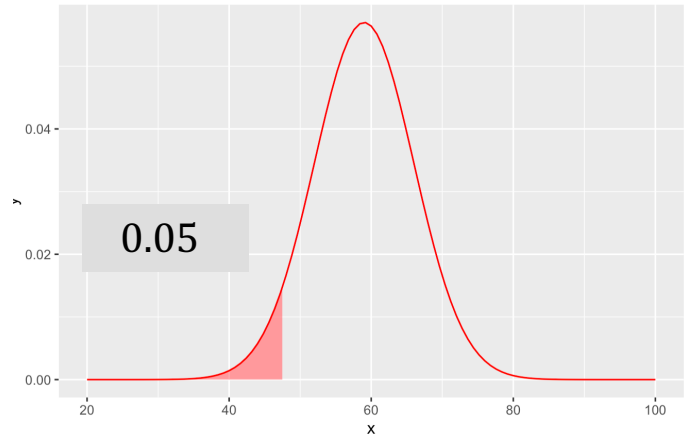
- our hypothesis should be ***one-tailed***

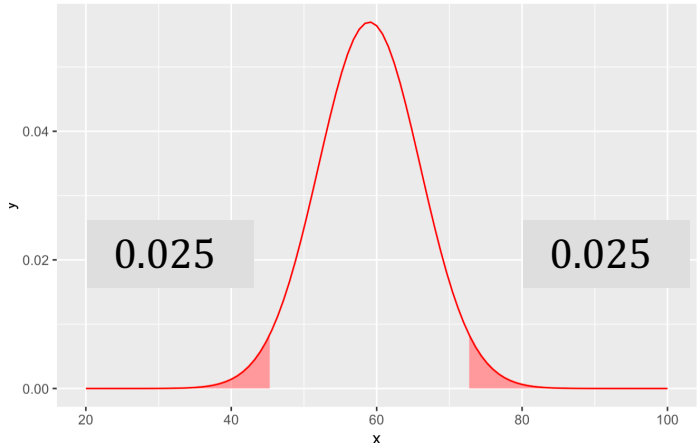
What if we wound someone tapping too fast ***also*** not to belong to the the healthy population either?

- If we think this may happen, our hypothesis should be ***two-tailed***
  - We would consider *extremely low* and *extremely high scores* as not being part of the population

What about our α?

- If we are ok with a 5% probability of Type I error, our test will need to reject the lowest 2.5% and the highest 2.5% (2.5% in each tail)





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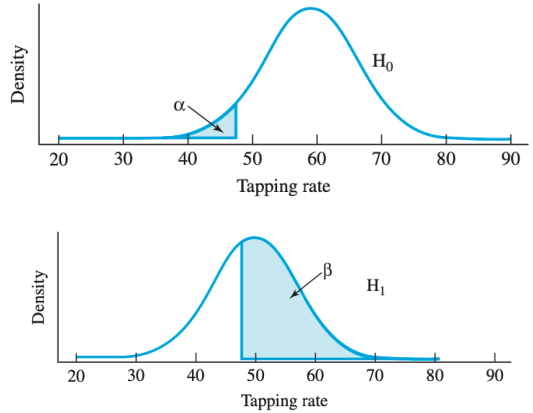
STATISTICAL POWER

	H_0 is true	H_0 is false
fail to reject H_0	True negative $1 - \alpha$	Type II error β
reject H_0	Type I error α	True positive $1 - \beta$



If H_0 is false

↗ β is the chance of not being able to reject H_0 when we should have

↗ $1 - \beta$ is the chance of being able to reject H_0 when we should have *(what we want!)*

$1 - \beta$ is the Statistical Power: chance at detecting an effect if there is one

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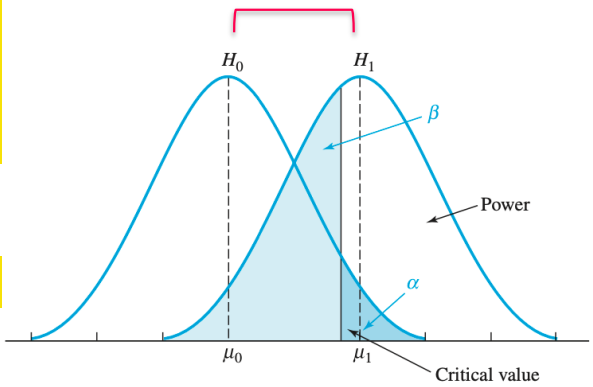
HOW DO WE INCREASE POWER?

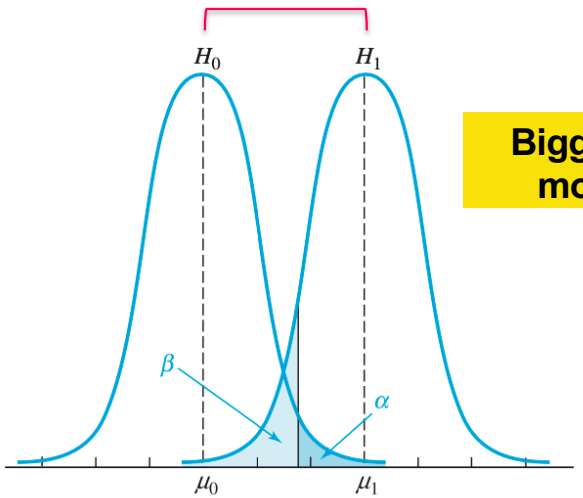
The more the distributions overlap, the higher β (chance of not rejecting H_0 when we should)

the less power we'll have

Effect size



Effect size



Bigger sample, more power

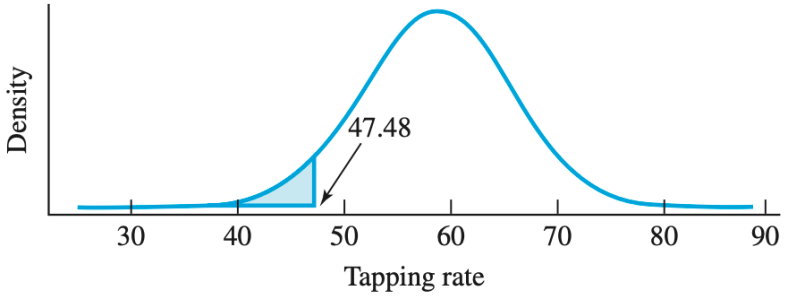
How do we increase power?

- ↗ Increasing α (but this comes with a higher probability of a Type I error)
- ↗ Increasing the effect size (distance between means) – usually beyond our control
- ↗ Increasing sample sizes (it will reduce variance, making the distribution more "pointy")

The **sample size** is crucial to get **sufficient statistical power** to detect an effect!

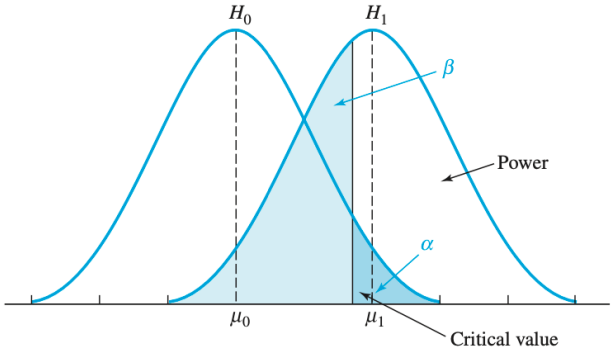
Questions?

SUMMING UP (1)



- In case of a **single score**,
  - We can compute **Probable Limits** indicating where scores may likely fall in a distribution
  - We can **reject** H_0 that the score comes from a population if this score is ***very unlikely*** to be drawn from the population
- In case of a **sample**,
  - We can compute **Confidence Intervals** where we expect to find the population parameter
  - We can **reject** H_0 that the sample is drawn from a population if the parameter is very unlikely to be drawn from the sampling distribution

SUMMING UP (2)



- Statistics comes with a process of decision making, trying to decide whether one of two hypotheses is true
- We set up a specific rejection level (e.g., 5%) and reject H_0 if the probability of the obtained result given H_0 is less than that rejection level
  - We make a Type I error when we reject H_0 although it is true (α: probability of Type I error)
  - We make a Type II error when we do not reject H_0 although we should have (β: probability of Type I error)
- Sample size is important, as it increases power (the chances of finding a difference if there is one)

HOMEWORK

Source: xkcd

- Chapter 9, sections 9.4 and 9.5 of “Statistics for Linguists”
- Chapter 8, sections 8.7 to 8.4 and Chapter 12, sections 12.1 to 12.2 of Howell, “Fundamental Statistics for the Behavioral Sciences” (available on Moodle)

- Watch this videos:
  - “Confidence Intervals” (Crash Course Statistics) https://www.youtube.com/watch?v=yDEvXB6ApWc